
UNIT 4 GRAPHICAL REPRESENTATION OF FUNCTIONS

Sets, Relations and Functions

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4.0 INTRODUCTION

From the previous unit, we know that a function is a relationship between its domain (X, in Fig. 4.1a) and co-domain (Y, in fig. 4.1a) such that every element of the domain is related to exactly one element of the co-domain/range. On this ground, Fig. 4.1a does not represent a function (or functional relationship) on each of the following two grounds:

Graphical Representation of Functions

- The element 5 of the domain (X) is related to three elements, viz. 4, 5, 7 of the co-domain.
- each of the elements 3 and 4 of domain X is not related to any element of the co-domain Y.

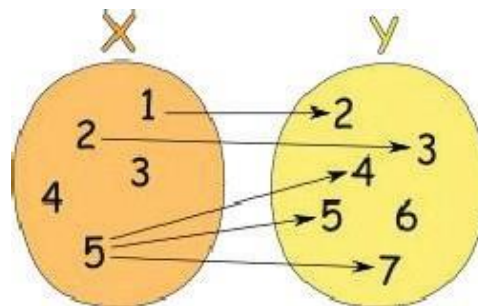


Fig.4.1 a: A Non-Functional Relationship

However, for a relationship to be a function, there is *no such restriction* on the elements of *co-domain* Y. For example, Fig. 4.1b represents a function or functional relationship, even when

- element C of co-domain Y corresponds to two elements viz. 2 and 3 of domain X, and
- element A (or even B) of co-domain Y is not related to any element of domain X.

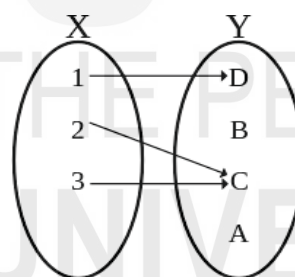


Fig.4.1 b: A Functional Relationship

4.1 OBJECTIVES

After going through this unit, you should be able to

- Apply vertical line test to identify a non-function;
- explain how co-ordinate geometry differs from the conventional/synthetic geometry;
- obtain algebraic equations corresponding to simple geometric figures;
- graphically represent various types of functions of a single variable; and
- handle the graphs of two variable functions at a preliminary level.

4.2 VERTICAL LINE TEST OF FUNCTION

The **vertical line test** for graphically testing *(only) failure* of a relationship $f(x)$ to be a function of x indicates that the vertical line crosses the graph of $f(x)$ at two distinct points (Fig 4.2).

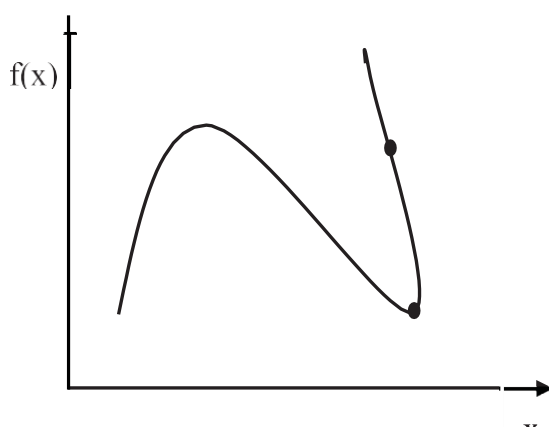


Fig. 4.2: Graph that does not Represent a Function

Thus, using vertical-line test we can identify the graphs that are not functions.

For example, you can show that each of the following graphs is not a function.

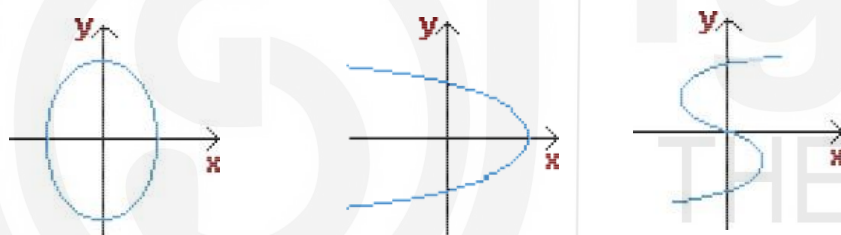


Fig.4.3: Graph of Not-Function

But the graph shown in Fig 4.4 below represents a function. Thus, some of the important equations such as ellipse, circle and parabola may not represent a function.

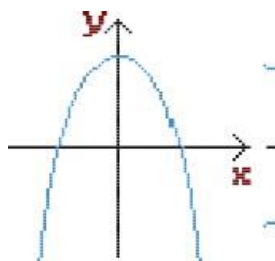


Fig.4.4: Graph of a Function

One must note that the above test can be used *to test whether a relationship is functional*. It cannot be used to test (or confirm) whether a relationship is functional. There are other additional requirements for a relationship to be functional.

4.2.1 Relationship between Equation and Function

Before we discuss functions in detail, we need to be clear about the possible relationship between an equation and a relation/function.

Consider the equation

$$X^2 + Y^2 = 4 \quad \dots\dots\dots (1)$$

Then to talk about mathematical relation/function, we rewrite the above equation so that one of the variables, say Y is written on L.H.S. and everything else is shifted to R.H.S. as follows:

$$Y^2 = 4 - X^2 \quad \dots\dots\dots (2)$$

Next step to talk about a relation/ function corresponding to Equation (1) is to rewrite (2), so that only Y (and not its power, or square-root etc.) is written on L.H.S. as follows:

$$Y = \pm \sqrt{4 - X^2} \quad \dots\dots\dots (3)$$

At this stage, after reducing the L.H.S. to a single variable, with the power of the variable as exactly 1, we can talk of a (mathematical) relation/function. As (3) does not satisfy single-value rule, hence, equation (1) does not correspond to a function. Of course, it corresponds to a relation. Equation (3) is obtained to make Y as dependent variable, and X as independent variable. By interchanging the roles of X and Y in the above process, we can make X as dependent variable, and Y as independent variable. Similarly, by taking the following equation in three variables

$$X^2 + Y^2 + Z^2 = 4 \quad \dots\dots\dots (4)$$

Through the similar process, we may rewrite the Equation (4)

$$\text{as: } Z = \pm \sqrt{4 - X^2 - Y^2} \quad \dots\dots\dots (5)$$

At this stage, we may talk of Z being a function or not, and this time, function of two independent variables, viz. X and Y. Again (5) does not represent a function.

4.2.2 Co-ordinate Geometry

Co-ordinate Geometry is an *approach* to investigating and solving problems in the domain of geometric objects such as straight lines, triangles, planes, cubes, etc. The ***approach consists in representing geometric objects as Algebraic expressions and equations*** like $ax + by + c = 0$ for a straight line, and $x^2 + y^2 = c$ for a circle etc. And then use algebraic tools for making inferences and solving problems. The process of translating geometric objects to algebraic expressions are achieved using ***co-ordinate systems***. There are several different co-ordinate systems, each more useful than the others for some specific purposes. We will discuss some of these briefly. However, before that we mention some synonymous and related terms, and a little bit of history.

Co-ordinate geometry is also called ***Analytical geometry***, or ***Cartesian geometry***

(in honor of *Rene Descartes*, 17th century French philosopher and mathematician). Though evidence of the use of co-ordinates in geometry is found in the work of earlier mathematicians, particularly some Greeks and the 11th-century Persian mathematician Omar Khayyam, yet the discipline in its current form was invented independently by two 17th-century mathematicians, viz. Rene Descartes and Pierre de Fermat.

Earlier forms of geometry, which have been practised for thousands of years, were retrospectively called, since the 17th century, synthetic geometry, which uses, instead of algebraic tools, the axiomatic method (based on proof techniques) to prove results from a few basic properties called postulates/axioms.

From our experience with school mathematics, we know that in 2-dimensional space, the Cartesian coordinate system uses two intersecting, mutually perpendicular real lines, as shown in Fig. 4.5.

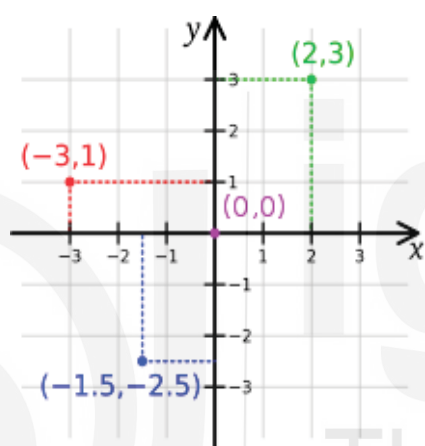


Fig 4.5: Two-Dimensional Cartesian Co-ordinate System

Though, we will be using mainly cartesian co-ordinate system, yet we should know the names of other co-ordinate systems which include **polar co-ordinate system**, **cylindrical & spherical co-ordinate system**, **homogeneous co-ordinate system** and **Curvilinear co-ordinate system**.

4.2.3 Cartesian Co-ordinate System

In the rest of the discussion in the unit, unless mentioned otherwise, the term 'co-ordinate system' will be used for cartesian co-ordinate system, which is also called rectilinear co-ordinate system.

In this co-ordinate system for 2-dimensional space, two mutually perpendicular intersecting lines (**called axes**), as shown in the Fig. 4.6 below, form the basis of *rectangular co-ordinate system for 2-dimension*

- 1) The horizontal number line is called the **x- axis**.
- 2) The vertical number line is called the **y-axis**.

The **origin** is where the two lines/axes intersect. (See Fig.4.6). It has four **Quadrants** marked with Roman numerals.

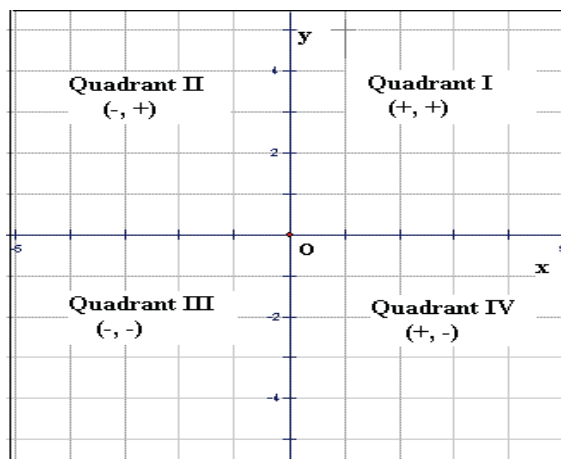


Fig. 4.6: Rectangular Co-ordinate System

Each point on the graph corresponds to an **ordered pair**. At the *origin*, it is assumed that value of $x = 0 =$ value of y . when dealing with an (x, y) graph, the **x co-ordinate** is always written first and the **y co-ordinate** as second in the ordered pair (x, y) .

Normally, the values of the **independent** variable (generally the **x -values**) are placed on the horizontal axis. In contrast, the values of the **dependent** variable (generally the **y -values**) are placed on the vertical axis.

For a point $P(x, y)$, the x -value, called the **abscissa**, is the perpendicular distance of P from the y -axis, and the y -value, called the **ordinate**, is the perpendicular distance of P from the x -axis.

The values of x and y together, written as (x, y) are called the **co-ordinates** of the point P . The centre of the co-ordinate system (where the two lines intersect) is called the **origin**.

Such a formulation is called the "**rectangular**" **co-ordinate system** because the scale used along the x -axis is evenly spaced, as is the scale along the y -axis.

4.3 TRANSLATING GEOMETRIC FIGURES TO ALGEBRAIC EQUATIONS

For translating a geometric figure, *say, a straight line*, we create a co-ordinate system around the object, or, equivalently, the geometric figure is embedded in a co-ordinate system as shown in Fig. 4.7. How and where to place the geometric figure in the co-ordinate system depends on the problem under consideration. Each point in Fig 4.7, has co-ordinates, as one of the points, say A , in the figure above, has co-ordinates $(2, 0)$, with 2 as abscissa and 0 as ordinate, and another point, say B , has co-ordinates $(0, -2)$.

To represent a geometric figure by an algebraic equation, we take an arbitrary point in Fig 4.7, assuming its co-ordinates as (x, y) with x as abscissa and y as ordinate. Then an **algebraic equation** representing the figure is some relationship (to be determined) between x and y .

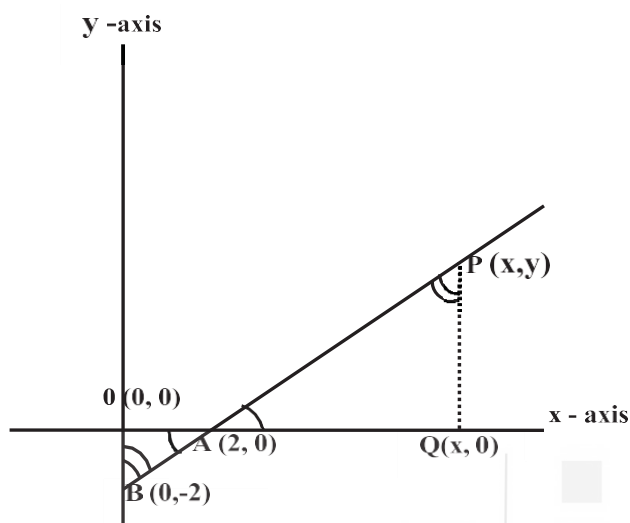


Fig. 4.7: Coordinate System

To derive the algebraic equation corresponding to Fig. 4.7, let us take $P(x, y)$ outside the segment AB , and drop a perpendicular PQ on x -axis. Then triangles AOB and AQP are similar. From conventional/synthetic geometry, we know:

$$|AO| / |AQ| = |BO| / |PQ|, \quad \dots (i)$$

Where, $|X|$ denotes length of line segment X .

Now

$$|AO| = 2, |AQ| = x-2, |BO| = 2, |PQ| = y \quad \dots (ii)$$

From (i) and (ii) we get $2 / (x-2) = 2/y$, implying

$y = x-2$ is the algebraic translation of the Figure 4.7.

Using similar method, we may obtain algebraic equations corresponding to the well-known curves, including the following standard algebraic forms

- $x^2 + y^2 = a^2$ for Circle

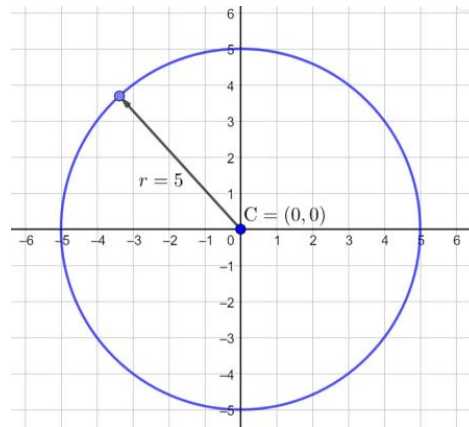


Fig. 4.8: Circle : $x^2 + y^2 = 5^2$

- $(x/a)^2 + (y/b)^2 = 1$ for Ellipse

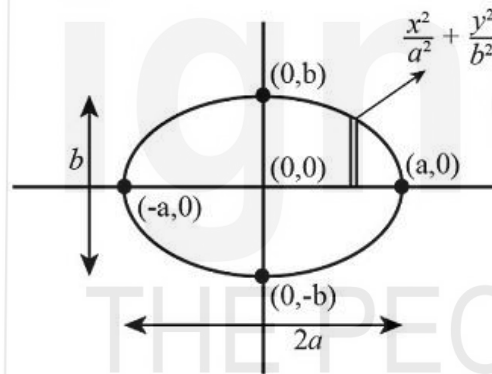


Fig. 4.9: Ellipse : $(x/a)^2 + (y/b)^2 = 1$

- $y^2 = 4ax$, with $a > 0$ for Parabola

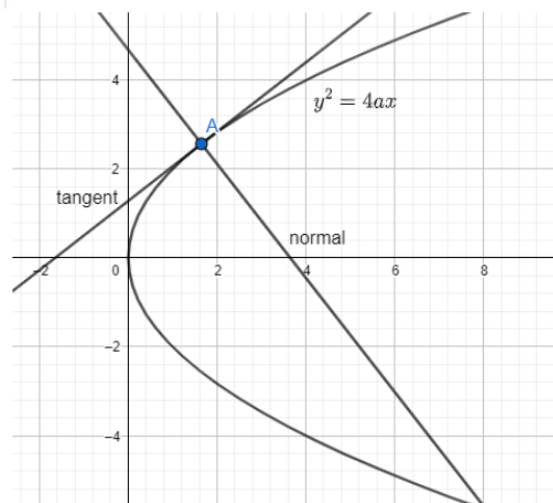


Fig. 4.10: Parabola : $y^2 = 4ax$, with $a > 0$

- $(x)^2 - (y)^2 = 1$ for Hyperbola

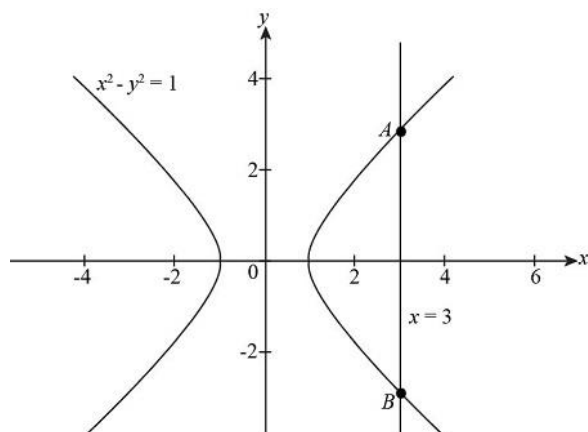


Fig. 4.11 : Hyperbola : $(x)^2 - (y)^2 = 1$

- $xy = c^2$ for Rectangular Hyperbola.

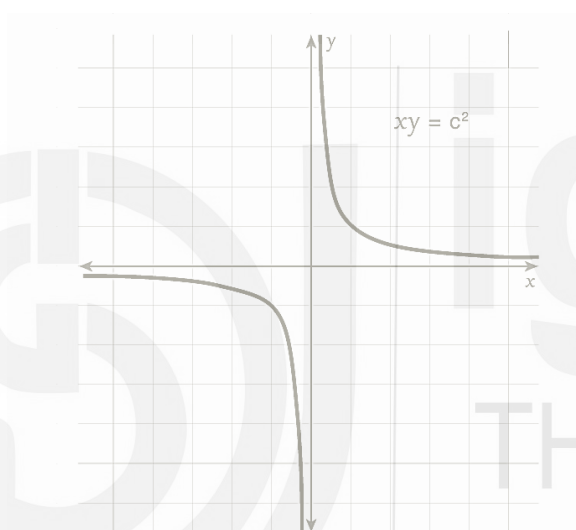


Fig. 4.12 : Rectangular Hyperbola : $xy = c^2$

4.4 GRAPHING LINEAR FUNCTIONS

A function $y = f(x)$, is called **linear** if the power of x in $f(x)$ is not more than 1. A linear function corresponds to a linear equation $Ax + By + C = 0$, in which degrees of x and y do not exceed 1, and A, B, C are constants. Where at least one of $A, B \neq 0$.

To plot a graph of a function, we take, one by one, values in the domain (x_i) as the independent variable x , and then find the corresponding values in the range y_i of the dependent variable y . It helps record the x and y values while plotting the graphs. For example, consider a linear function depicting the relationship between price (x) and market demand (y) for a commodity. To plot the graph for evaluating the relationship, it is convenient to first prepare a table, called a t -chart, comprising the values of x and y . If a function takes the form of $y = 7 - 5x$, then t -chart is as under:

Table 4.1: t-chart of $y=7-5x$

x	$y=7-5x$
-1	12
0	7
1	2
2	-3
3	-8

In general, we may need to take many values of the independent variable x and find the corresponding values of y . But in the *case of functions*, we need to take **only two** values of the independent variable x and calculate two corresponding values of the dependent variable y . Thus, to draw the required graph, we **need only** the following part of the table :

X	$Y = 7-5X$
-1	12
0	7

The graph of such a function is as follows (refer Fig. 4.13):

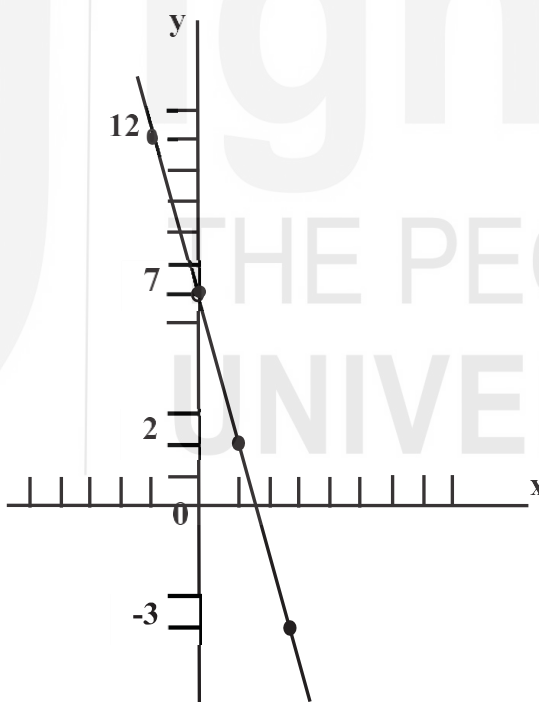


Fig.4.13:Graph of Function $y=7 - 5x$

The graph has ***x-intercept*** on a point where y is zero and ***y-intercept*** on a point where x is zero. For example, to find the x and y intercepts of the equation $3x+4y=12$ proceed as follows:

To find the x -intercept, set $y=0$ and solve for x . That is, $3x+4(0) = 12$ or, $x = 12/3=4$. Similarly, to find the y -intercept, set $x = 0$ and solve for y .

That is, $3(0) + 4y=12$ or, $y=12/4=3$. Thus, the x -intercept is $(4,0)$ and the y -

intercept is (0,3).

As a linear equation represents a straight line, and a straight line is specified by only two of its point. Hence, the graph of the line is as shown in Fig. 4.14.

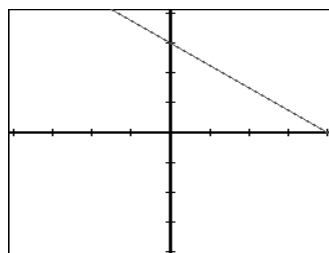


Fig.4.14: Graph of $3x+4y=12$

4.4.1 Absolute Value Function

Absolute Value Function is given as $f(x) = |x|$ indicating that we need to consider modulus $|x|$ of a real number keeping it as the non-negative value without regard to its sign. That is, $|x| = x$ for a positive x , $|x| = -x$ for a negative x (in which case $-x$ is positive), and $|0| = 0$.

Thus, the function $y = |x|$, is defined by the following two equations

$$y = |x| = x \text{ for } x \geq 0, \quad \dots(1)$$

$$\text{and } y = |x| = -x \text{ for } x < 0 \quad \dots(2)$$

As each of (1) and (2) is a linear equation, each is represented by a straight line as shown in Fig. 4.15 below:

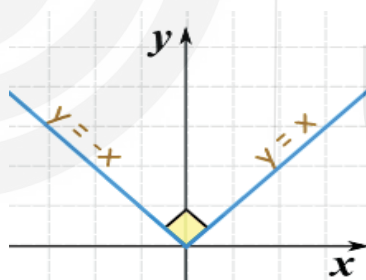


Fig.4.15: Graph of Absolute Value

However, for some functions, the x-intercept & y-intercept method may not yield the required result, because the required curve may not intersect x-axis or y-axis.

Example: f is a function given by $f(x) = |x| + 2$. The y intercept is given by $(0, f(0)) = (0, |-2|) = (0, 2)$; x intercept is at the point $(2, 0)$ since we solve for $|x-2| = 0$. But $0 = y = f(x) = |x-2| = 0$ has no solution.

Since $|x-2|$ is either positive or zero for $x=2$; the domain of f is the set of all real numbers and the range of f is given by the interval $[0, +\infty)$. The corresponding graph is shown in Fig 4.16.

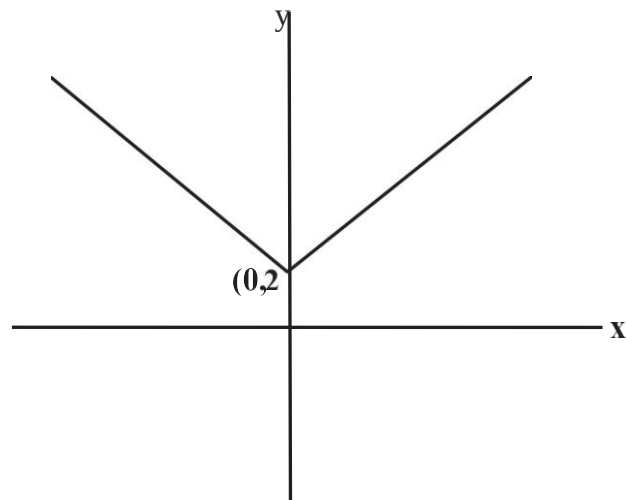


Fig.4.16: Graph of $f(x)=|x|+2$.

In general, for drawing a graph for a linear function, we need **just two pairs** (x_i, y_i) $i = 1, 2$ of values of (x, y) .

4.4.2 Step Function

A step function (or staircase function) is a piecewise function containing all constant “pieces” . The constant pieces are observed across the adjacent intervals of the function, as they change value from one interval to the next. A step function is discontinuous (not continuous) as you can see in the Fig. 4.17. You cannot draw a step function without removing your pencil from your paper.

Example of a step function: For definition of our particular step function $y=f(x)$,

First, we write every $x = \text{Int} - x + \text{Fraction-}x$, where $\text{Fraction-}x \geq 0$.

E.g., for $x = 5.46$, $\text{Int-}x = 5$, and $\text{Fraction-}x = .46$

For $x = - 3.87$, $\text{Int-}x = -4$, and $\text{Fraction-}x = .13$

Then we define $y = f(x) = \text{Int-}x$

Thus, $f(5.46) = 5$, and $f(-3.87) = -4$.

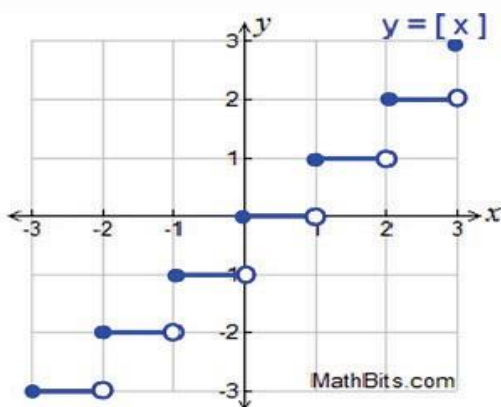


Fig.4.17: Graph of Step Function

Note: The *domain* of a function is the set of all inputs for which the function is defined. For piece wise functions, the domain is union of all the individual cases. The *range* of a function is the set of all the possible function outputs. For piecewise functions, this is the union of the ranges, so fall back to the individual cases.

Graphical
Representation
of Function

Check your Progress -1

- 1) Draw the graph of the equation $y^2 = x+5$ and comment if it is a function.

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- 2) Find algebraic equation corresponding to a circle of radius r.

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- 3) Draw the graph of $y = (-5/3)x - 2$.

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- 4) Find when $f(x)=2|3x+2|-12$ crosses the x-axis.

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5) A truck rental company, charges a rental fee of Rs. 25 plus Rs. 0.30 per mile for driving between 0 and 500 miles for the standard –sized truck. For the same truck, the company charges Rs. 100 plus Rs. 0.15 per mile for driving more than 500 miles, but less than 1000 miles. Write a step function that represents the situation.

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4.5 GRAPHING NON-LINEAR (CURVE TYPE) FUNCTIONS

Determining the nature of the functions that produces non-linear type curves requires additional considerations compared to those of linear types. When you graphed straight lines, you only needed two points to graph the line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly *not* enough/sufficient for graphing curves of quadratic or cubic functions. We discuss the caution to be exercised for sketching the graphs of such function.

4.5.1 Even and Odd Functions

A function f is **even** if its graph is symmetric with respect to the y - axis. This criterion can be stated algebraically as follows: f is even if $f(-x)=f(x)$ for all x in the domain off. For example, if you evaluate at 3 and at-3, then you will get the same value, if f is even. Geometrically, such functions are symmetric about the y -axis.

We have seen that for an even function $f(x)=f(-x)$ for all x , there is symmetry about y -axis. The graph, in this case is shown in Fig 4.18.

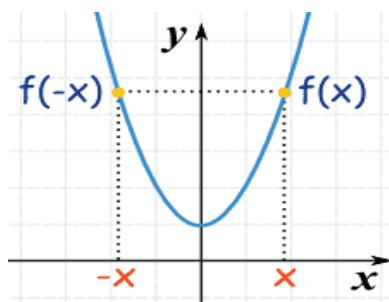
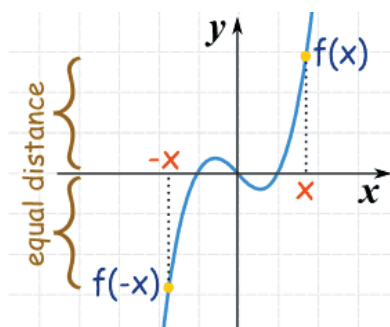


Fig.4.18: Graph of Even Function

A function is "**odd**" when $f(x) = -f(-x)$ for all x and symmetry is observed from the point of origin (Fig. 4.19).

**Fig. 4.19: Graph of Odd Function**

Note: The graph based on $f(x)=x^3-x$ is an odd function. It is symmetrical about the origin.

4.5.2 Quadratic Functions

Quadratic functions help describing demand, cost, revenue and profit which you will find while covering microeconomic analysis. The general technique for graphing quadratics is the same as **for graphing linear functions**. However, since quadratic graphs are represented by curvy lines (for example, parabolas, ellipses etc.), rather than the straight lines generated by line equations, there are some additional considerations.

The most basic quadratic is $y = x^2$. When you graphed straight lines, you only needed two points to graph your line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly not be enough points for graphing a quadratic. For example, compute these three points:

Table 4.2: T-chart of $y=x^2$

x	$y=x^2$
0	0
1	1
2	4

Then, based only on this experiment, the plotted graph will produce a straight line. Thus, the graph is not the correct presentation of the function. More points need to be considered. By extending the above table with more points we construct Table 4.3.

Table 4.3: Extension with more points

X	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

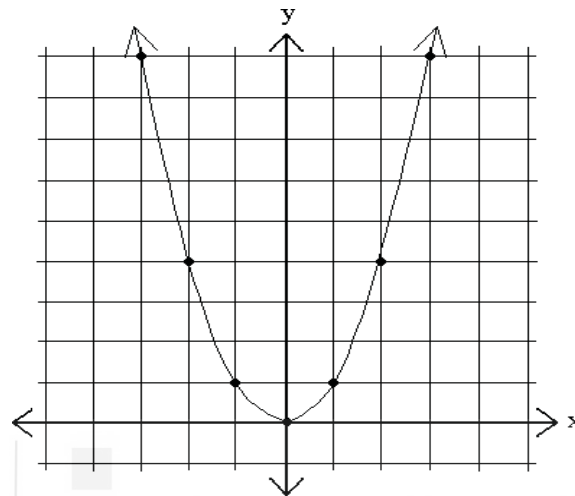


Fig. 4.20: Graph of Equation $y=x^2$

We observe that the graph in Fig 4.20 is a parabola. Note that the parabola does not have a constant slope. In fact, as x increases by 1, starting with $x=0$, y increases by 1,3,5,7,... As x decreases by 1, starting with $x=0$, y again, increases by 1,3,5,7,....

Consider a quadratic function of the form: $f(x) = ax^2 + bx + c$ with $a \neq 0$. The graph of such a function takes one of the two general forms shown in Fig. 4.21, depending on the sign of a .

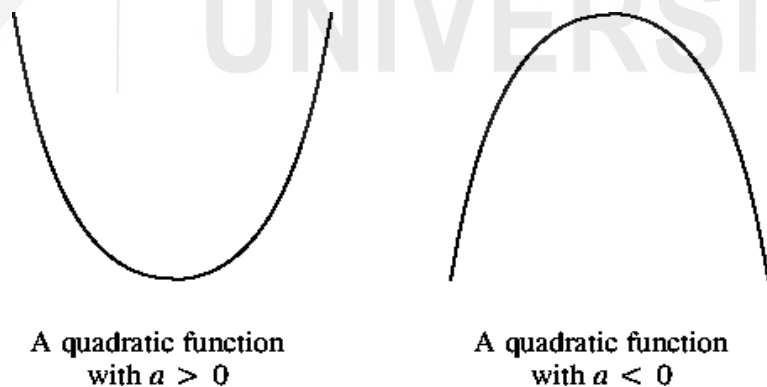


Fig.4.21: Graph of Quadratic Function

Example: Sketch the graph of function $y = 2x^2 - 8x + 6$. As the coefficient of x^2 is positive, so the graph is *U-Shaped* (Fig.4.22)

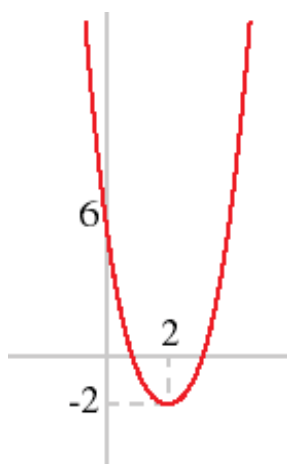


Fig.4.22: Graph of Function $y=2x^2-8x+6$

In the graph of $y = x^2$, the point $(0, 0)$ is called the vertex. The vertex is the lowest point in a parabola that opens upward. In a parabola that opens downward, the vertex is the highest point. We can also graph a parabola with a different vertex. For example, see the graph of $y=x^2+3$ (Fig 4.23). The graph is shifted up 3 units from the graph of $y = x^2$, and the vertex is $(0, 3)$.

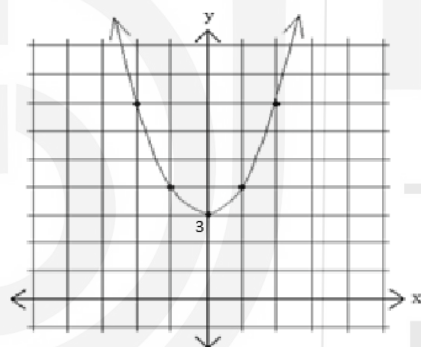


Fig. 4.23: Graph of $y=x^2 + 3$

We can also shift the vertex left and right. Observe the graph of $y=(x+3)^2$ in Fig 4.24.

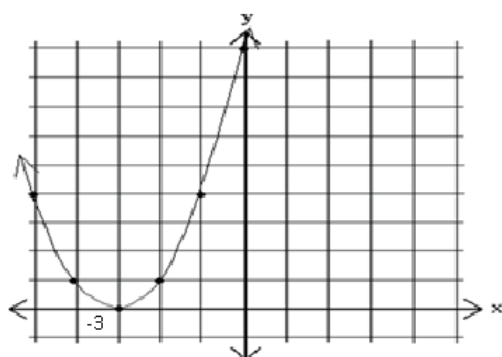


Fig.4.24: Graph of $y=(x+3)^2$

The graph is shifted *left* 3 units from the graph of $y=x^2$, and the vertex is $(-3, 0)$. Next, see Fig. 4.25, the graph of $y=(x-3)^2$, which is obtained by

shifting to the right.

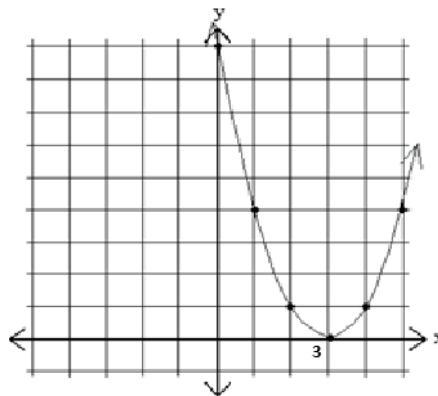


Fig. 25 : Graph of $y = (x-3)^2$

The graph is shifted to the *right* 3 units from the graph of $y=x^2$, and the vertex is (3, 0). Remember that the **axis of symmetry** is the line which divides the parabola into two symmetrical halves. It is given by the equation

$x=h$. For example, the line of symmetry in the graph (Fig. 4.26) of $y = (x-2)^2 + 1$ is $x = 2$.

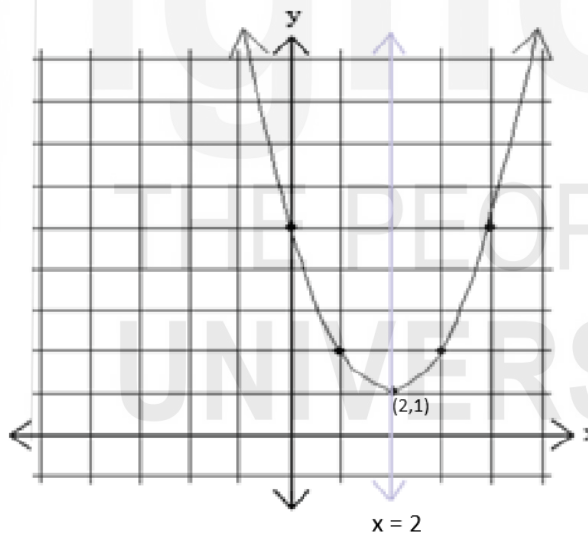


Fig.4.26: Axis of Symmetry

4.5.3 Computation of the co-ordinates of Vertex

Consider a generic quadratic expression: $y=ax^2+bx+c$. Start with **completing the square** on the equation to have $y = a \left[x^2 + \frac{bx}{a} + \frac{c}{a} \right]$ or $y=a \left[\left(x + \frac{b}{2a} \right)^2 - \left(\frac{b}{2a} \right)^2 + \frac{c}{a} \right]$

The expression $-\left(\frac{b}{2a} \right)^2 + \frac{c}{a}$ is a constant and it does not depend on x . So, we can replace it with k .

Thus, we write :

$$y = a \left[\left(x + \frac{b}{2a} \right)^2 + k \right]$$

Now, depending on whether a is positive or negative, the parabola given by y will either have either a maximum value or a minimum value. Since a and k are fixed, this must occur when $\left(x + \frac{b}{2a} \right)^2 = 0$. Hence, $x = -b/2a$ which implies that the function y has either a minimum value or a maximum value when $x = -b/2a$.

Thus, x -coordinate of the vertex is $-b/2a$. Y -coordinate can be obtained by replacing, in the equation $y = ax^2 + bx + c$, the variable x by $(-b/2a)$.

Example: Find the vertex of $y = 3x^2 + x - 2$ and graph the parabola.

To find the vertex, we look at the coefficients a , b and c . Then k is obtained by evaluating y at $h = -\frac{1}{6}$.

$$x\text{-coordinate of the vertex} = \left(-\frac{b}{2a} \right) = \left(-\frac{1}{2 \times 3} \right) = -\frac{1}{6}$$

The y -coordinate is given by replacing x by $-\frac{1}{6}$ in the equation

$$y = 3x^2 + x - 2.$$

$$= 3(-1/6)^2 + (-1/6) - 2 = 3/36 - 1/6 - 2$$

$$= 1/12 - 2/12 - 24/12 = -25/12$$

We get y -coordinate as $-\frac{25}{12}$. Thus, the vertex is at $(-\frac{1}{6}, -\frac{25}{12})$. T-chart can be prepared as shown in Table 4.4.

Table 4.4: T-chart of $y = 3x^2 + x - 2$

x	$y = 3x^2 + x - 2$
-2	8
-1	0
0	-2
1	2
2	12

The graph (Fig. 4.27) can be drawn with the specification vertex:

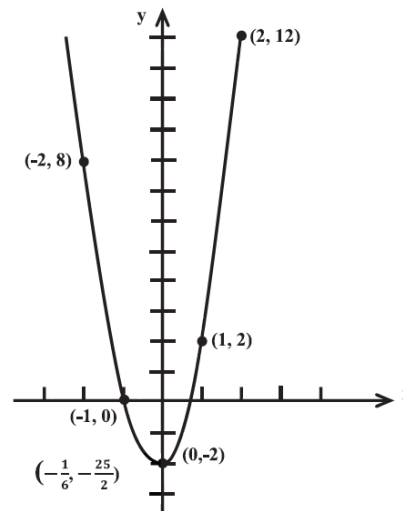


Fig.4.27: Graph of $y=3x^2+x-2$

4.5.4 Cubic Functions

Cubic functions are encountered in the discussion of total cost in microeconomic analysis. Consider a cubic function of the form $f(x) = ax^3 + bx^2 + cx + d$. Its graph (Fig 4.28) is obtained as follows:

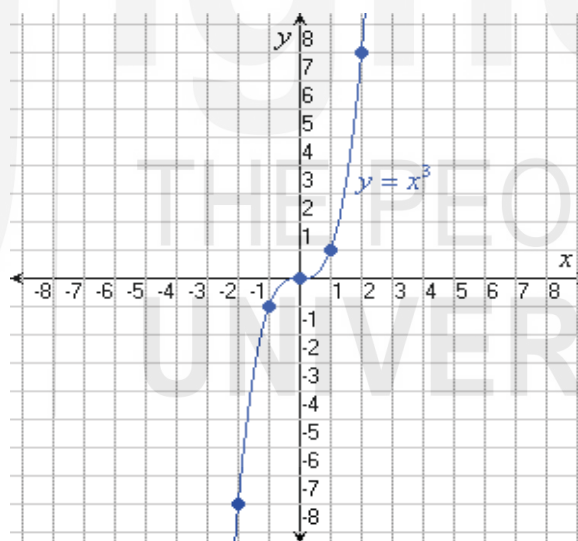


Fig. 4.28: Graph of Cubic Function $f(x) = x^3$

To trace the cubic function given by $f(x) = x^3$, find the x and y intercepts of the graph of f , find the domain and range of f and then sketch the graph of f . The y intercept is given by $(0, f(0)) = (0, 0)$. The x co-ordinates of the x -intercepts are the solutions to $x^3 = 0$. The one x -intercept is at the points $(0, 0)$. The domain of $f(x)$ is the set of all real numbers. Since the leading coefficient of x^3 is positive, the graph of: f is up on the right and down on the left and hence the range of: f is the set of all real numbers. Table of values is as follows:

x	-2	-1	0	1	8
$f(x)=x^3$	-8	-1	0	1	8

Since $f(-x) = -f(x)$, function f is odd and its graph is symmetric with respect to the origin $(0,0)$.

4.6 GRAPHING FUNCTIONS ASYMPTOTIC TYPES

We are familiar with exponential growth models such as population growth and compound interest. Such models are represented by functions, which are called asymptotic functions. Next, we examine behavior of such functions through their graphical representations.

4.6.1 Graph of Asymptote

An asymptote is, essentially, a line that a graph approaches, but does not intersect (Fig. 4.29). For example, in the following graph of $y=1/x$, the line approaches the x -axis ($y=0$), but never touches it. No matter how far we go towards infinity, the line will not actually reach $y=0$, but will always get closer and closer to it.

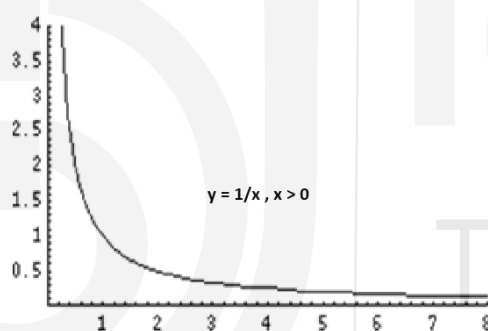


Fig. 4.29: Graph of Asymptote

This means that the line $y=0$ is a horizontal asymptote. Horizontal asymptotes occur most often when the function is a fraction where the top remains positive,

but the bottom goes to infinity. Going back to the previous example, $y=1/x$ is a fraction: When we go out to infinity on the x -axis, the top of the fraction remains 1, but the bottom gets bigger and bigger. As a result, the entire fraction gets smaller, although it will not approach zero. As we increase the values of x , the values of the function will gradually have values $1/2$, then $1/3$, then $1/10$, even $1/10000$, but never quite 0. Thus, $y=0$ is a horizontal asymptote for **the function $y=1/x$.**

The function also serves as an example of **vertical asymptote, because as y becomes larger and larger in absolute value, x becomes smaller and smaller in magnitude, but never becomes 0.**

Similarly, the equation $y = \frac{4}{x-2}$, gives one of the horizontal asymptotes, viz. $y=0$, and one of the vertical asymptotes, viz. $x=2$ (Fig.4.30).

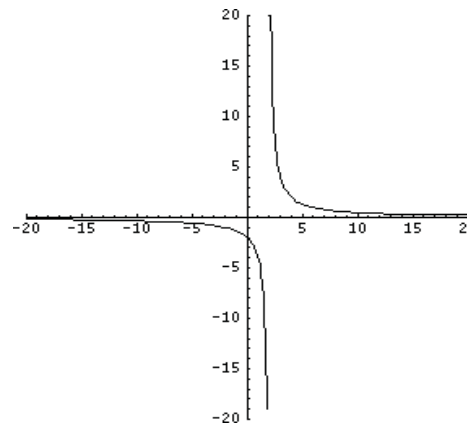


Fig. 4.30: Graph of the Function $y = \frac{4}{x-2}$

Note that as the graph approaches $x=2$ from the left, the curve drops rapidly towards negative infinity. This is because the **number** **or** is staying at 4, and the denominator is approaching to 0. That means that the fraction itself is getting very big and negative. When x is exactly 2, the function does not exist because you cannot divide by 0. Immediately after 2 it resumes at positive infinity, because the numerator is 4 and the denominator is again very tiny, but this time it is positive.

Next, we consider two examples of functions one of which gives only

- (i) horizontal asymptote, but not a vertical asymptote, and (ii) vertical asymptote, which is not horizontal.
- i) The function $y=1/(x-2)$, for $0 \leq x < 2$, is asymptotic with $x=2$, being a vertical asymptote. Similarly,
- ii) The function $x=1/(y-2)$, for $0 \leq y < 2$ is asymptotic with $y=2$, being a horizontal asymptote

4.6.2 Square Root Function

As the name indicates we take a variable x to write the function in the form of $f(x)=\sqrt{x}$, with Domain/Range: $[0,\infty)$, and intercept $(0,0)$, and which is neither even nor odd and is increasing on $(0,\infty)$. Plotting its graph yields Fig. 4.31.

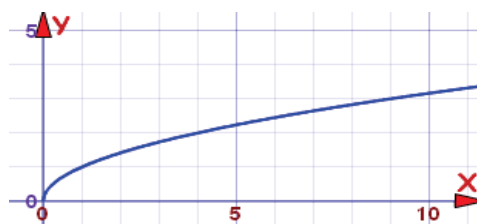


Fig.4.31: Graph of Square-root Function

You may see that square root function is used to depict an *is o cline* which provides information on the way different *o* quant are related to each other. In fact, this *o* cline based on square root function converges to a point where output is maximized and marginal product is zero.

4.6.3 Exponential Function

A general exponential function is of form $y=a^x$, where $a > 0$, $a \neq 1$, constant, and x any real variable. The discussion of the general exponential function requires substantial space, and hence, will not be discussed here. We discuss a special such function which is found useful in several applications, viz. the natural exponential, which is of the form $y=e^x$ where e is Euler's number (a number approximately equal to 2.718281828). If we take its domain as the set of all real numbers i.e., $(-\infty, \infty)$, then its range is $(0, \infty)$. Since the graph never intersects or goes below the x -axis, they-values are not zero or negative (Fig. 4.32).

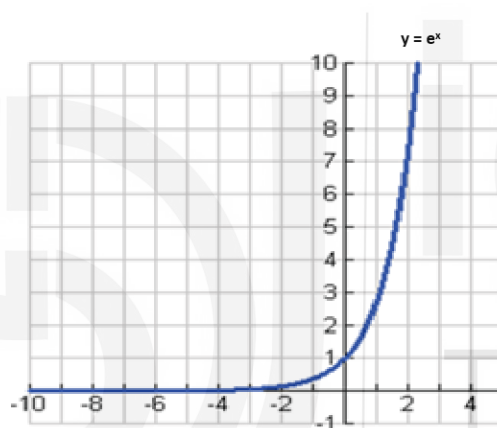


Fig. 4.32: graph of Exponential Function

Three of the most common application of exponential and logarithmic functions (to be discussed soon) have to do with interest earned on an investment, population growth, and carbon dating. For example, exponential functions are used to describe growth or decay processes with constant growth (or decay rate), population growth, growth of GDP, inflation etc.

4.6.4 Logarithmic Function

A logarithm function is a sort of inverse function of exponential function. Thus, if $b > 0$, $b \neq 1$, $y = b^x = f(x)$ is an exponential function, then $\log_b y = g(y)$ (read as log of y to the base b) is such that, with some restrictions, $f(g(y)) = y$, and $g(f(x)) = x$. A function given by $y = \log_b x$, where b is any number such that $b > 0$, $b \neq 1$, and $x > 0$, denotes logarithmic function. When $b > 1$, the values of the logarithm increase; and for $0 < b < 1$, the values decrease.

The logarithmic function slowly grows towards positive infinity as the variable increases and slowly goes towards negative infinity as it approaches 0. Three special cases of logarithm are quite well-known and useful: (i) $b=10$, called decimal or common logarithm and is more used in science and engineering

applications. (ii) $b = e$, approximately equal to 2.718281828 is called natural logarithm and used frequently in Mathematics and physics, and (iii) base $= 2$, is called binary logarithm, and is frequently used in computer science. For all bases b , including the three mentioned here, $\log b = 0$

In the logarithmic function, the domain is all positive real numbers (never zero) while the range is all real numbers. The graph is asymptotic to the y-axis - gets very, very close to the y-axis but does not touch it or cross it (Fig 4.33).

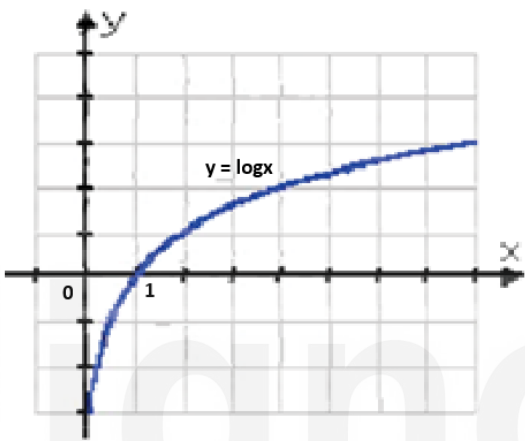


Fig. 4.33: Graph of Logarithmic Function

Check Your Progress -2

- 1) Draw the graph for the function, $y = x^2 - 2x$.
.....
.....
.....
- 2) Let f be a rational function defined by $f(x) = \frac{x+1}{x-1}$. Find the x and y intercepts of the graphs of the function.
.....
.....
.....
- 3) Draw the graph of an average fixed cost curve. Why do you think it is asymptotic?
.....
.....
.....
-

4.7 GRAPHING FUNCTIONS WITH TWO VARIABLES

In the above discussion we have seen that for a function of one variable, $y=f(x)$, the graph of f is the set of all points (x, y) in 2-dimensional space such that $y=f(x)$. In general, these points lie on a curve in the plane. Now consider a function of two variables.

The graph of a function of two variables is a *surface in three dimensions*. Thus we deal with a 3D (**three-dimensional graph**). In geometry and physics, **3D** stands for **three-dimensional**. It describes a space or object that has three measurements: **length**, **width**, and **depth** (or height).

While a 2D shape (like a square on paper) is flat, a 3D object (like a cube) has "volume"—it takes up physical space. Though discussion on 3D is not in the scope of this course, here you are introduced to 3D graphs for the sake of completeness. Further, if you need to understand more about this topic, you may explore it on your own.

To define a point in 3D space, we use a co-ordinate system with three axes, usually labeled as x , y , and z .

- X-axis (Length): Left to right.
- Y-axis (Height): Up and down.
- Z-axis (Depth): Forwards and backwards.

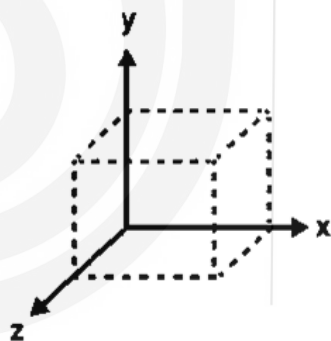


Fig. 4.34 : 3D Coordinate System

The graph of a function $z = f(x, y)$ of two variables, or the graph of a relationship $g(x, y, z)$ among three variables. For example, here (Fig. 4.35) is the graph of $z=2x^2+2y^2-4$.

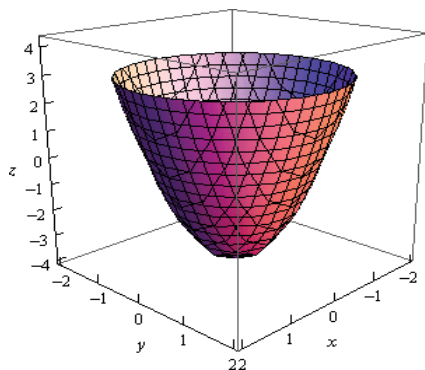


Fig. 4.35: Three-Dimensional Graph

The information in a three-dimensional graph is often collapsed into a two-dimensional graph with the use of contour lines. The procedure is: The x and y axes are retained, but instead of depicting z or $f(x,y)$ axis, all x, y combinations giving rise to the same z or $f(x, y)$ values are connected with a contour line; an arbitrary number of these may be shown for various values of z of $f(x,y)$.

4.6.4 Level Curves

Let f be a function of two variables, and c is a constant. The set of pairs (x, y) such that $f(x, y)=c$ is called the **level curve** (or, contour lines) of f for the value c .

Example: Let $f(x, y)=x^2+y^2$ for all (x, y) . The level curve of f for Value 1 is the set of all pairs (x, y) such that $x^2+y^2=1$, a circle of radius 1. This set is shown in Fig. 4.36.

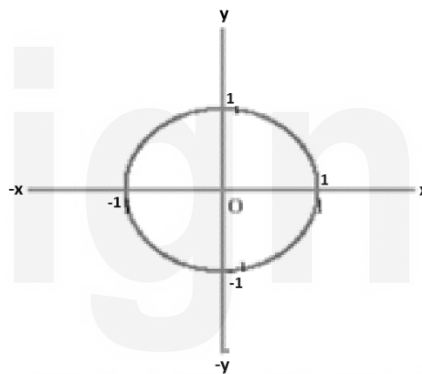


Fig.4.36: Level curve

Economists call the level curves of a utility function *indifference curves* and those of a production function *isoquants*. Thus, level curves have algebraic equations of the form $f(x, y) = k$ for all possible values of k . A contour diagram or contour map of a function $f(x, y)$ is a 2-dimensional graph (Fig. 4.37) showing several level curves in the xy -plane corresponding to several values of k .

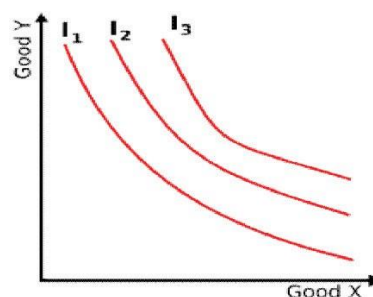


Fig.4.37: Indifference Curve Map

Example: Cobb-Douglas Production Model

Given P =the monetary value of all goods produced, K = total capital investment over the same period, and L = the total labor force, the following relationship between these quantities holds: $P = 1.01L^{0.75}K^{0.25}$

The contour diagram shown in Fig. 4.38 below shows that reducing labour increases capital investment in order to maintain the same level of production i.e., supposedly more investment is needed in machinery to make up for the lack of available labour. Conversely, increasing labour reduces capital investment since supposedly the work is being done by the labor force.

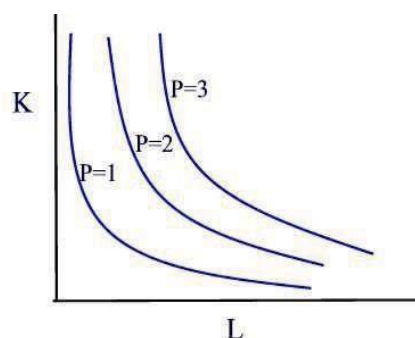


Fig. 4.38: Indifference Curve Map- Cobb Douglas Example

Example:

Let $f(x,y)=x^2-y^2$. We will study the level curves $c=x^2-y^2$.

First, look at the case $c=0$. The level curve equation $x^2 - y^2 = 0$ factors to $(x - y) \cdot (x + y) = 0$. This equation is satisfied if either $y = x$ or $y = -x$. Both these are equations for lines, so the level curve for $c=0$ is two lines.

See that this is the equation for a hyperbola. If c is positive, the hyperbolas open to the left and right. If c is negative, the hyperbolas open up and down.

For example, if $c=1$, the equation is $x^2-y^2=1$. If $c=-1$, the equation is $x^2-y^2=-1$. Several level curves are plotted in Fig. 4.39.

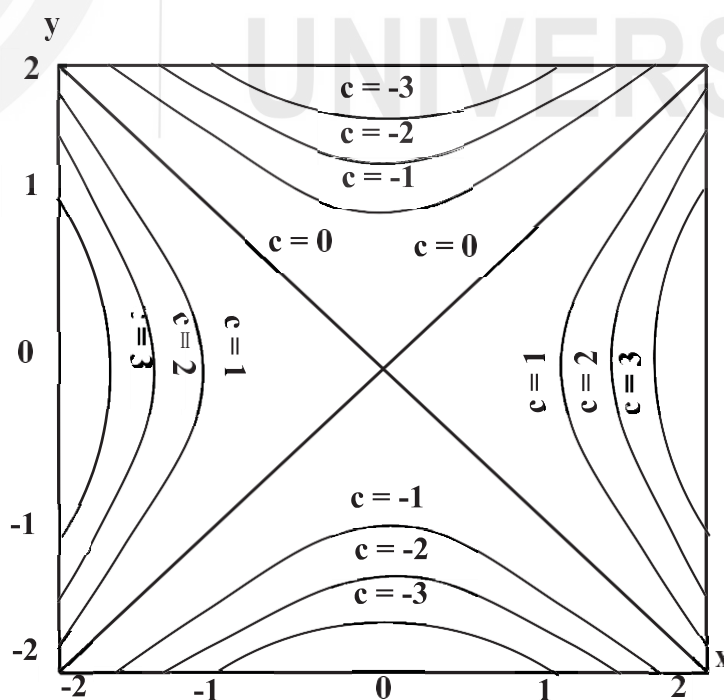


Fig.4.39: Level Curves

Check Your Progress -3

1) Explain the meaning of level curve.

.....

.....

.....

2) Let $f(x,y) = xy$ for all (x,y) . Show a collection of level curves of f .

.....

.....

.....

3) Suppose a function is defined by $f(x,y)=1$ for all (x,y) . Comment on the shape of level curves.

.....

.....

.....

4.8 SUMMARY

In this unit, we mainly discussed the method of sketching graphs of functions. However, first we discussed how to obtain algebraic equations for well-known geometric figures. In the reverse direction, for the purpose of graphing functions, starting with the relationship between elements of domain and co-domain, we recalled the basis of functional relationships. An operational method for dealing with a function was presented to identify dependent and independent variables. Before proceeding with the process of sketching the graph of a function, the use of the vertical line test to determine whether a relationship is a functional relationship was discussed. According to this test, the equations of circles, ellipses, etc., fail to meet its requirement of a functional relationship.

While discussing the method of graphing, this unit considered functions in five groups, viz., linear, curve, asymptotic, piecewise, and curve with branches. Linear graphs were drawn when the functions were linear, absolute value, or step-like. In cases of graphs showing curves, functions included were odd and even, quadratic and cubic. Among the graph of asymptotic types, functions included were square root, exponential and logarithmic. Graphs appearing in

pieces in the xy-plane comprised rational functions, piecewise functions and discontinuous functions. Finally, in this unit, in the last group of graphs shaving *curve* with branch functions concerned with hyperbola and rectangular hyperbola were discussed.

4.9 KEYWORDS

Abscissa	:	x-coordinate (horizontal) of a point m a 2-dimensional co-ordinate system.
Absolute Value Function	:	Obtaining the <i>absolute value</i> (or <i>magnitude</i>) of a number.
Asymptote	:	A line that a curve approaches, as it heads towards infinity.
Asymptotic	:	Graph approaching a point but never touching the Discontinuity point.
Axis of Symmetry	:	A line of symmetry for a graph. The two sides of a graph on either side of the axis of symmetry look like mirror images of each other.
Cobb-Douglas Production Function	:	Production function which is applied to describe how much output two inputs into a production process make.
Contour Diagram	:	Graph that provides one way of visualizing (contour map) functions of two variables.
Co-ordinates	:	Any of a set of two or more numbers used to determine the position of a point, line, curve, or plane in a space of a given dimension with respect to a system of lines or other fixed references.
Cubic Function	:	A polynomial function of degree 3.
Directrix	:	A line perpendicular to the axis of symmetry of a parabola.
Discontinuous:		A function which for certain values or between certain values of the variable does not vary continuously as the variable increases.
Even Function:		University function is $f(x)$ is said to be even provided that $f(x)=f(-x)$.
Exponential Function	:	Function defined by $ex\ p(z)=e^z$.
Hyperbola	:	A hyperbola is a curve where the distances of any point from: a fixed point (the focus), and a fixed straight line (the directrix) are always in the same ratio.
Indifference Curve	:	A group of indifference curves for two commodities

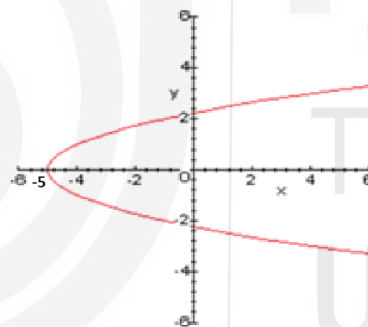
Graphical Representation of Functions

Level Curves	:	A contour diagram or contour map of a function $f(x,y)$ is a 2 dimensional graph showing several level curves in the xy -plane.
Logarithmic function	:	A function (as $y = \log_a x$ or $y = \ln x$) that is the inverse of an exponential function (as $y = a^x$ or $y = e^x$) so that the independent variable appears in a logarithm.
Odd Function	:	Univariate function $f(x)$ is said to be odd provided that $f(-x) = -f(x)$.
Piecewise Function	:	A function can be in Pieces. (split function)
Point Discontinuity	:	Graph of the function has a hole at $x=a$.
Quadratic Function	:	A function that can be described by an equation of the form $(x)=ax^2+bx+c$, where $a \neq 0$.
Rational Function	:	A function that is the ratio of two polynomials.

4.10 ANSWERTO CHECK YOUR PROGRESS

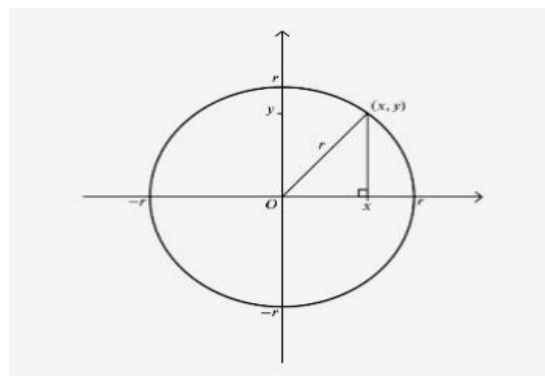
Check Your Progress -1

1) The graph is



By the vertical line test, this graph is not the graph of a function, because There are many vertical lines that intertsects the graph more than once.

2) As in the case of straight line, for translating the geometric figure a circle, create a co-ordinate system around the object, with centre of circle as origin of co-ordinate system, as shown below.



Mark an arbitrary point, say P, on the circle, as (x, y). Draw a perpendicular from P to X-axis, meeting it in say point Q. Then length of perpendicular is of length y. and x is the length of the intercept on X-axis. Then from conventional/ synthetic geometry, we know

$$x^2 + y^2 = r^2$$

is an algebraic equation of the circle with radius r.

- 3) First prepare the t-chart which is Table :t-chart of $y = (-5/3)x - 2$

x	$Y = (-5/3)x - 2$
-6	8
-3	-2
0	2
3	-7

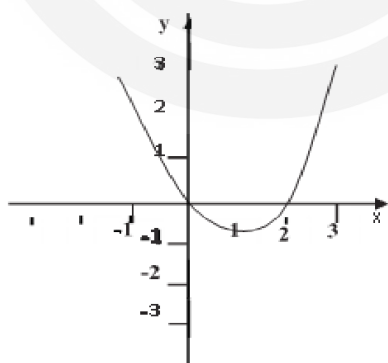
Then you draw the graph.

- 4) $x = 4/3$ or $x = -8/3$

- 5) $f(x) = \begin{cases} 25 + 0.30x, & \text{for } 0 < x \leq 500 \\ 100 + 0.15x, & \text{for } 500 < x < 1000 \end{cases}$

Check Your Progress-2

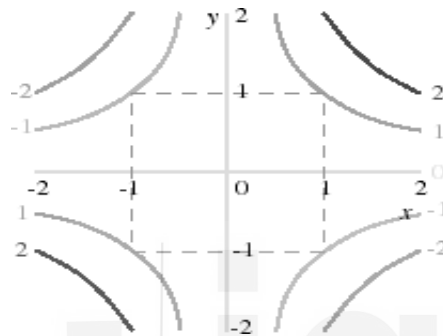
- 1)



- 2) The x intercept is found by solving $f(x) = 0$ or, $x + 1 = 0$. The x intercept is at the point $(-1, 0)$. The y intercept is at the point $(0, f(0)) = (0, -1)$.
- 3) Since total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases without t becoming zero.

Check Your Progress -3

- 1) A curve in two dimensions on which the value of a function $f(x, y)$ is a constant.
- 2) The following figure may be seen. The number beside each curve is the value of the function to which it corresponds. The level curve for the value 0 consists of the axes.



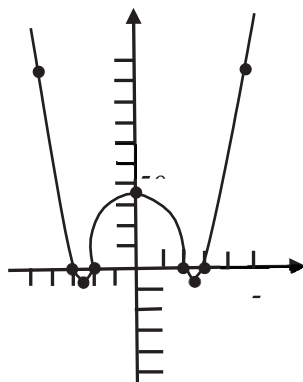
- 3) A level curve is defined as a *set*, and indeed some level curves of some functions are not "curves" at all. In the given function the value of f is 1 for all values of x and y . As a result, you get straight lines, not curves. Moreover, the level curve of this function for the value 2 is empty as there are *no* values of (x,y) such that $f(x,y)=2$.

4.11 SOME MORE EXERCISES

Q1. Let $y = x^4 - 13x^2 + 36$. Draw its graph.

Ans. This is a positive even power ("to the fourth"), so the graph will be upon both ends. Factoring the polynomial, we get $y = (x+3)(x-3)(x+2)(x-2)$, so the zeroes (x -intercepts) are -3, -2, 2, and 3. If you plot a few other points on your T-chart, it will be no trouble to graph this:

x	$y = x^4 - 13x^2 + 36$
-4	84
-2.5	-6.19
-1	24
0	36
1	24
2.5	-6.19
4	84



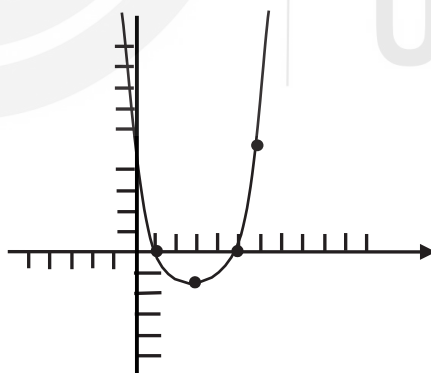
Graph of $y=x^4-13x^2+36$

Q2. A quadratic function is given as $y=x^2-6x+5$. Draw its graph.

Ans. We need to plot more than three points to get the correct shape. Find the intercepts, which in this case are at (1, 0), (5, 0), and (0, 5); find the vertex, which in this case is at (3, -4). Remember that parabolas look like "smilies", not like straight lines. So, we want to plot a few more points:

Table: Chart of $y=x^2+6x+5$

x	$y=x^2+6x+5$
0	5
1	0
3	-4
5	0
6	5



Graph of $y=x^2+6x+5$

4.12 REFFERENCES AND FURTHER READINGS

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